# Machine Learning for a Multi-Sensor Digital Twin

Gait, activity, place and events from seven days of self-monitoring

Participant: Hemel

Scripts: `pipeline/ml_gait.py`, `ml_heelstrike.py`, `ml_activity.py`, `ml_place.py`, `ml_fusion.py`, `ml_events.py`, `ml_figures_a.py`

# 1 Summary

The seven days of multi-sensor data were used to build and test the machine-learning layer of a *digital twin*: a minute-by-minute estimate of how the participant walks, what he is doing, where he is and whether something unusual happened. Every model was tested on days it had never seen (leave-one-day-out), trained only on labels that came from a different source than its inputs, compared with a simple baseline, and kept small enough to run on a phone. The main results are:

- **Gait timing from the foot IMU alone.** A temporal convolutional network (TCN, 32k parameters) finds heel strikes in the raw motion stream with F1 0.915 (precision 0.93, recall 0.90) and a timing error of $4.5 \pm 8.6$ ms on unseen days, better than a tuned peak detector on all seven days (F1 0.898). It transfers to the other foot (F1 0.89–0.91). For toe-off, a gradient-boosting model and a CNN were no better than the existing IMU rule (stride-to-stride scatter 40 vs 39 ms): the pressure-sensor reference itself shifts by about $\pm 40$ ms between sessions, so the rule is already at the noise floor of its own labels.

- **Activity from the insoles** (walking, still, vehicle): macro-F1 0.79 with gradient boosting on engineered features; walking and still are recognised with F1 0.91 each. One insole alone still gives 0.74–0.75, so the twin survives an insole dropout. Adding the pressure sensors made it worse (0.66) because they changed physically during the study. A smooth metro ride is invisible to the feet (0 of 73 windows detected).

- **Place from the air** (bedroom, indoor, outdoor, vehicle): macro-F1 0.67 from the environment node alone, mainly from sound level, humidity and $CO_2$; bedroom F1 0.96, indoor 0.77, outdoor 0.80. Vehicles (49 minutes in total) could not be learned.

- **Fusion into twin states** (bedroom, indoor-still, outdoor-walking, vehicle): macro-F1 0.70, and vehicle recall rose from 0.14 (air) and 0.16 (feet) to 0.33 when both were combined with phone GPS.

- **Events without labels.** The participant's slip ranked first of 18,261 steps by impact relative to the walk's own typical step, though the margin to the next step is small. A generic anomaly detector instead highlights walking transitions (starts, stops, stairs, boarding a rickshaw). All seven noted exposure events (restaurant, market, smoke, soldering fumes, two metro rides, metro station) were detected as episodes, and

their fingerprints separate combustion smoke, solder flux fume and crowded-vehicle $CO_2$.

- **Phone budget.** All neural networks are 32–50k parameters (32–49 kB as 8-bit) and run in 0.2–2 ms per window on one CPU core; the place model is a 4 kB logistic regression.

Two lessons run through all tasks: *label quality mattered more than model choice*, and *physical knowledge (two feet alternate, a smooth train does not shake the feet, fume particles are sub-micron) beat extra model complexity.* With one participant and seven days, simple models generalised better than neural networks except where data were plentiful (heel strikes).

# 2 The digital twin and the machine-learning tasks

The twin is organised in layers, each answering one question from one group of sensors (Table 1). Layers are combined at the end, and each layer is designed to cover a weakness of another: the activity layer tells the gait layer when not to count steps (vehicles), and the place layer sees the metro ride the feet cannot feel.

Table 1: Layers of the digital twin and the machine-learning task behind each.

| Task | Question | Sensors | Output |
|---|---|---|---|
| A | How am I walking? | Foot IMU (both insoles) | Heel strike and toe-off of every stride; stance and swing |
| C | What am I doing? | Foot IMU (+ phone GPS) | Walking / still / vehicle every 5 s |
| B | Where am I? | Environment node | Bedroom / indoor / outdoor / vehicle every minute |
| Fusion | Twin state | B + C + GPS | Bedroom / indoor-still / outdoor-walking / vehicle |
| E | Did something unusual happen? | Foot IMU; environment node | Slip candidates; exposure episodes with fingerprint |

# 3 Data and general method

## Data

Seven days (8–14 October 2026) of one participant: two pressure–IMU insoles (112 samples/s; heel, first and fifth metatarsal pressure sensors; orientation quaternion and acceleration), an environment node ($CO_2$, $PM_1$, $PM_{2.5}$, $PM_{10}$, temperature, humidity, sound; 10-s averages), phone GPS, a wrist band (heart rate, 1-min steps, sleep) and the participant's notes on events (rides, stairs, lift, restaurant, market, smoke, soldering, slip). The resulting datasets are 15,186 strides (11,115 with a valid pressure toe-off), 18,766 labelled heel-strike impacts in 48 continuous IMU streams, 6,776 activity windows, 4,286 environment minutes and 18,261 detected steps.

## Validation: why leave-one-day-out

Neighbouring strides, windows and minutes are nearly identical. Randomly splitting them between training and test sets would let a model recognise a near-copy of the test sample and greatly overstate its accuracy. Every result in this report therefore comes from *leave-one-day-out* testing: train on six days, test on the seventh, repeat for every day, and pool the seven test days. Where relevant, models were also trained on one foot and tested on the other, the closest available stand-in for a new wearer or a differently placed insole.

## Labels: independent of the inputs

A model must not be judged against labels made from its own inputs, or it simply re-learns the labelling rule.

- Gait (A): labels from the pressure sensors (toe-off) or from the impact search window defined by them (heel strike); inputs from the IMU only.
- Activity (C): "walking" = minute with $\geq 80$ steps on the *wrist band*, "still" = 0 band steps, "vehicle" = noted ride; inputs from the insoles only.
- Place (B): the participant's session contexts and notes; inputs from the environment node only.
- Events (E): no labels used for detection; the participant's notes only for checking.

### Baselines, models and metrics

Each task has a simple baseline (a rule, a peak detector or thresholds) that the models must beat, a classic model on engineered features (gradient boosting or logistic regression) and a small neural network on the raw signal, so that the choice is made by evidence. Classification is reported as precision (share of detections that are right), recall (share of true cases found), F1 (their harmonic mean) and macro-F1 (the average F1 over classes, so that small classes count as much as large ones). Timing is reported in milliseconds. A "sticky" Viterbi smoothing step, which assumes activities and places last longer than one window, is applied where stated.

### Designed for a phone and for a second participant

Inputs avoid anything tied to this participant's body or home: foot tilt is measured from the foot's own flat position found from the IMU (not from the pressure sensors, and independent of how the IMU is mounted), pressure is normalised per sensor, and the environment features include trends as well as levels. Models are small and work on streaming windows. A second participant (a woman) will be recorded; the same code then allows leave-one-person-out tests.

## 4 A. Gait layer: heel strike and toe-off from the foot IMU

### IMU-only signals

For each insole the orientation quaternion gives the direction of gravity in the sensor frame. The foot's "flat" orientation is the median gravity direction over the stillest samples (lowest rotation speed and acceleration closest to $1\,g$), which in walking recordings are mid-stance moments. Foot tilt is the angle from that orientation; tilt rate, rotation speed and acceleration magnitude complete the four channels. The earlier pipeline found the flat position with the pressure sensors; the IMU-only version gave the same toe-off results and removes the last dependence on them.

## Toe-off: can a model beat the rule?

Each stride (impact to impact) was resampled to 100 points per channel. The target was the pressure toe-off as a percentage of the stride, available on right-foot days 1–3 (4,153 strides) and left-foot days 3–7 (6,962 strides). Three methods were compared: the hand-made rule (toe-off when the push-off rotation rate has fallen to half its peak), gradient boosting on 101 features (curves sampled every 5 % plus timing landmarks) and a 1D CNN (34k parameters).

Table 2: Toe-off error on unseen days (ms). Within-session scatter is the stride-to-stride error after removing each session's mean offset.

| | Right foot (4,153) | | | Left foot (6,962) | | |
|---|---|---|---|---|---|---|
| | Rule | Boosting | CNN | Rule | Boosting | CNN |
| Mean absolute error | 31.6 | 41.0 | 42.2 | 88.9 | 54.7 | 55.5 |
| Bias | −12.6 | +10.2 | +10.1 | −84.2 | −6.8 | −8.1 |
| Within-session SD | 39.3 | 40.1 | 40.8 | 54.4 | 52.4 | 54.8 |
| MAE, other foot | – | 69.3 | 68.9 | – | 69.5 | 66.5 |

None of the models improved the stride-to-stride precision (Table 2). Figure 1 shows why. The error is dominated by *session offsets*: on the right foot each session's pressure toe-off sits between −42 and +11 ms from the IMU toe-off, on the left foot between −35 and −120 ms. The offset depends on how the insole and its sensors sat that day, which nothing in the foot's motion can predict. On the left foot the models appear better only because they learned the left insole's typical lag; trained on the other foot, the same models were wrong by about 60 ms. The conclusion for the thesis is that the IMU rule is at the noise floor of its pressure reference, and that the pressure toe-off is not a gold standard.

## Heel strike from the raw IMU stream

The pipeline located each heel-strike impact with the help of the pressure sensors. A phone-only twin must find strides in the raw IMU stream. Each stream was resampled to 100 Hz and a temporal convolutional network (dilated convolutions covering about 1.3 s of context, 32k parameters) was trained to output, for every sample, the probability that a heel strike happens there; peaks above 0.4 at least 0.6 s apart are detections. Training used windows from labelled walking regions (where every step is known) and from the noted

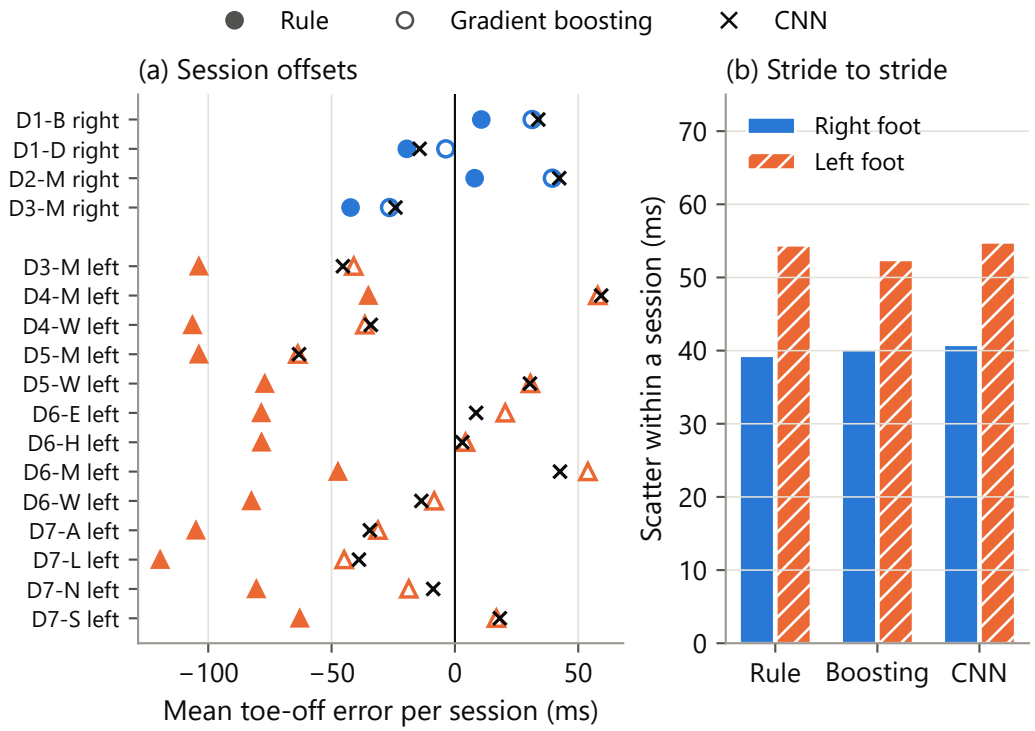



Figure 1: Toe-off from the IMU against the pressure-sensor toe-off, on unseen days. (a) Mean error of each recording session for the rule (filled), gradient boosting (open) and the CNN (crosses); the spread between sessions is the shifting pressure reference. (b) Stride-to-stride scatter within a session: the three methods are equally precise.

rides and lift (where there are no steps). The baseline is a peak detector on acceleration magnitude whose threshold was tuned on the training days.

The TCN beat the peak detector on every test day (Table 3, Fig. 2), mainly by finding about 900 more steps (recall 0.92 vs 0.87 before the two-foot check), and it transferred across feet. The remaining errors are largely in the labels: missed impacts had median peaks of 18 $\mathrm{m/s^2}$ against 40 for detected ones (slow, turning or shuffling steps where "the impact" is ill-defined), and a third of the false alarms were within 100 ms of a label.

Vehicles were the weak spot: rickshaw and metro vibration produced 10–19 false steps per minute. A physical constraint halved this without loss of accuracy: in a vehicle a bump reaches both feet at the same instant, whereas walking heel strikes alternate about half a stride apart. A detection is kept only if the other foot has no detection within ±150 ms and one 0.25–1.0 s

Table 3: Heel-strike detection on unseen days (pooled; a detection within ±50 ms of a labelled impact is a hit; with the two-foot check).

| | TCN | Peak detector |
|---|---|---|
| Precision | 0.934 | 0.945 |
| Recall | 0.896 | 0.856 |
| F1 | **0.915** | 0.898 |
| F1 per test day (range) | 0.877–0.979 | 0.854–0.964 |
| Timing error (mean ± SD) | $4.5 \pm 8.6$ ms | $3.1 \pm 6.9$ ms |
| F1, trained on the other foot | 0.89–0.91 | 0.90 |
| Vehicle false steps/min, one foot | 19.3 | 10.1 |
| Vehicle false steps/min, two-foot check | 10.6 | 6.3 |

away. The remaining false steps are removed in the twin by the activity layer, which switches step counting off in vehicles.

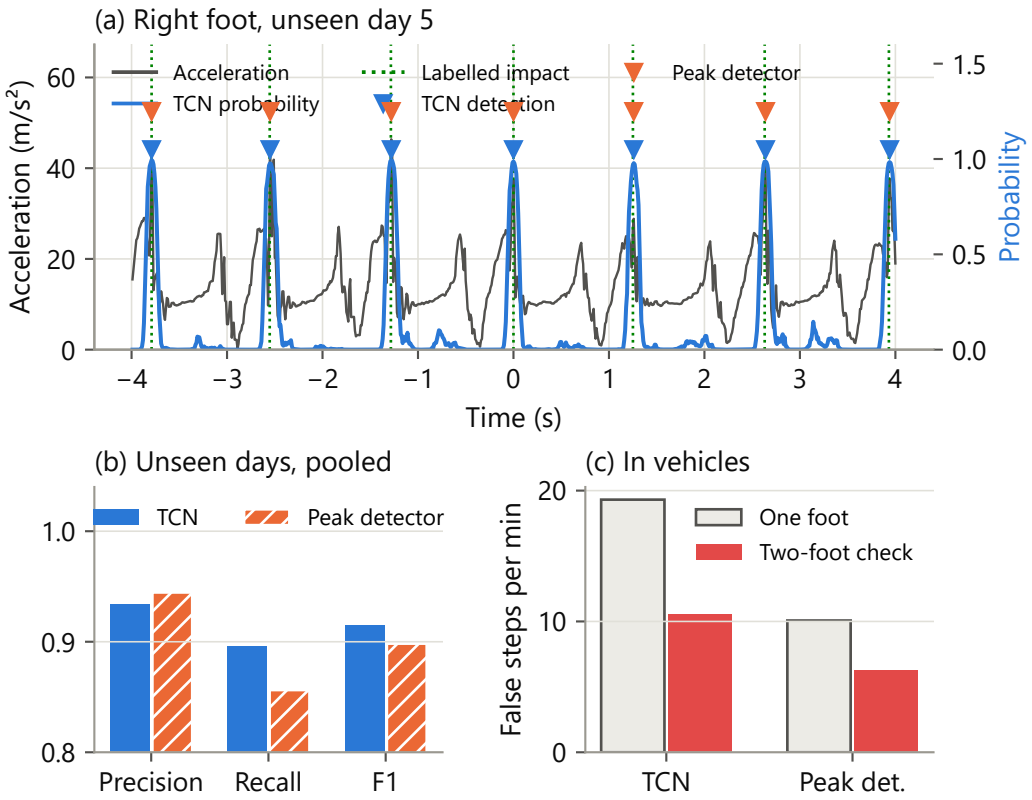



Figure 2: Heel-strike detection from the IMU stream. (a) Eight seconds of the right foot on day 5, a day the TCN never saw: acceleration, the TCN's heel-strike probability, labelled impacts (dotted) and the detections of the TCN and the peak detector. (b) Pooled precision, recall and F1 on unseen days. (c) False steps per minute in vehicles with one foot and with the two-foot alternation check.

# 5 C. Activity layer: walking, still or in a vehicle

## Set-up

Ten-second windows (every 5 s) were cut from all insole streams, aligned between the feet. Per foot, 15 features describe acceleration level and spread, how the acceleration energy splits between the walking band (0.5–3 Hz) and the vibration band (> 8 Hz), stride periodicity (autocorrelation at 0.7–2 s), foot-tilt range, rotation speed and the number of acceleration peaks; three two-foot features measure whether the feet move at the same instant or

alternately. Labels: 2,321 walking, 3,017 still and 280 vehicle windows (two rickshaw rides, one other vehicle ride and a metro ride on days 6–7); 53 stairs and 19 lift windows were kept aside. An important correction during development: labelling whole "transport" sessions as vehicle wrongly included station walking; labelling only the noted rides raised macro-F1 from 0.71 to 0.79.

## Results

Table 4: Activity recognition on unseen days (both insoles, smoothed).

| | Gradient boosting (features) | | | CNN (raw signal, 50k parameters) | | |
|---|---|---|---|---|---|---|
| | Precision | Recall | F1 | Precision | Recall | F1 |
| Walking | 0.92 | 0.90 | 0.91 | 0.92 | 0.91 | 0.91 |
| Still | 0.88 | 0.94 | 0.91 | 0.89 | 0.91 | 0.90 |
| Vehicle | 0.89 | 0.39 | 0.54 | 0.35 | 0.29 | 0.32 |
| Macro-F1 | | **0.79** | | | 0.71 | |

- *Walking and still* are recognised reliably by both models (F1 0.90–0.91; Table 4, Fig. 3).
- *Engineered features beat the CNN* overall (0.79 vs 0.71), entirely through vehicles. With about 5,600 labelled windows and only 280 vehicle windows, frequency-band features carry knowledge that the CNN would have to learn from very few examples.
- *One insole is enough* for walking and still: right only 0.74, left only 0.75 (Fig. 4). The CNN, trained with one foot randomly blanked, reached 0.75–0.77 with one foot.
- *Pressure sensors hurt* (0.66): they were replaced or failed during the study, so their behaviour on one day did not hold on another. This supports an IMU-based twin.
- *Phone GPS speed* helped vehicles slightly (recall 0.39 → 0.44) at a cost in precision; GPS is missing indoors and underground.
- *Smoothing* added 0.01–0.06 macro-F1 (nothing for the pressure model).

- *Case studies.* All 53 stairs windows were classified as walking, which is physically right (a barometer would be needed to tell stairs apart). The lift was classified as still (14 of 19 windows) or vehicle (5).

Vehicles are hard for a physical reason (Fig. 5). The day-7 rickshaw (foot vibration SD 3.3 $m/s^2$) was detected in 89 % of its windows and the day-6 vehicle ride in 67 %, but the gentler day-6 rickshaw (1.7 $m/s^2$) in only 22 %, and the metro (0.2 $m/s^2$, close to sitting still at 0.1) in none: in a smooth train the feet are genuinely still. With rides on only two days, each test day also had to learn from a different kind of ride.

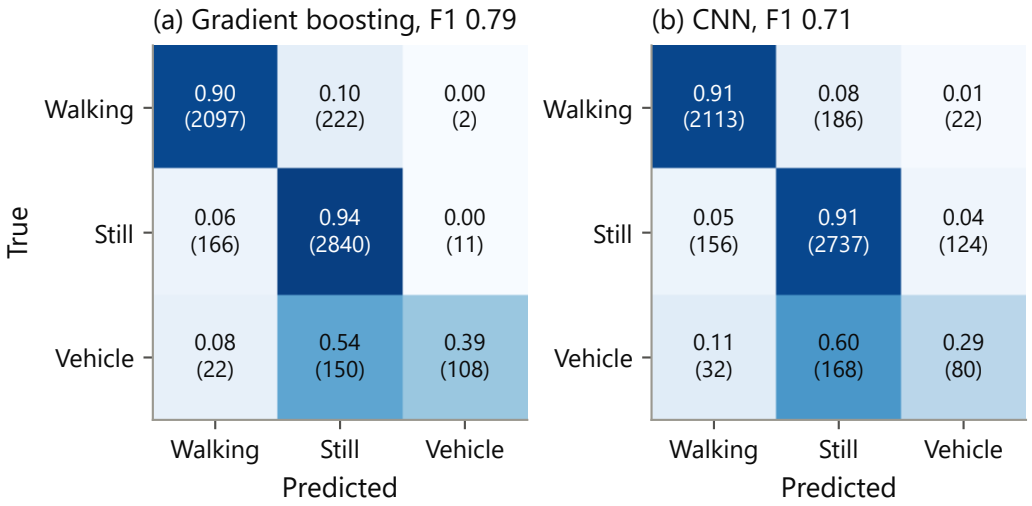



Figure 3: Activity recognition on unseen days (both insoles, smoothed): share of each true class (rows) assigned to each predicted class, with window counts. (a) Gradient boosting on engineered features; (b) CNN on raw 10-s windows.

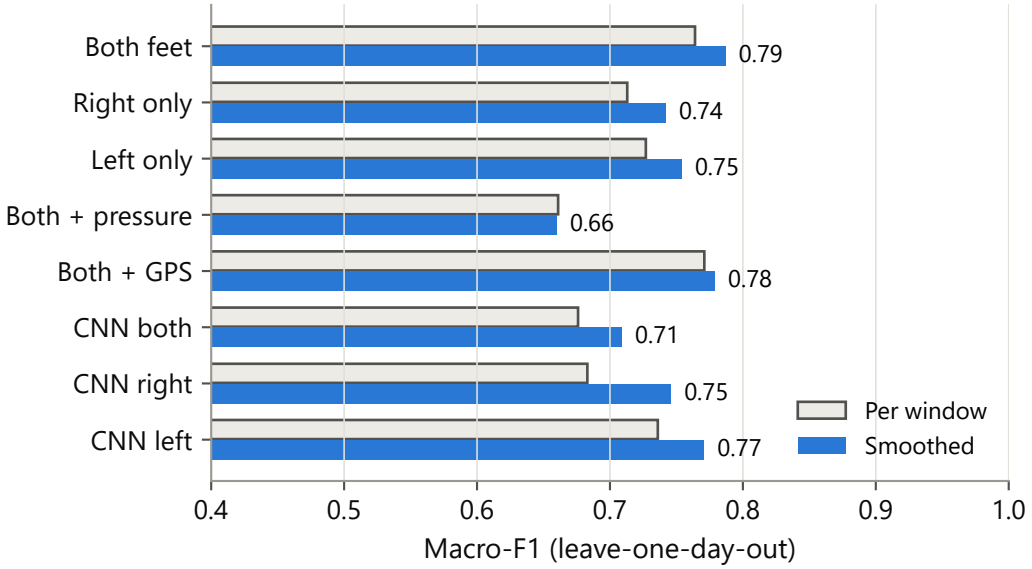



Figure 4: Effect of inputs and model on activity recognition (macro-F1 on unseen days, per window and after smoothing). "Pressure" adds the insoles' pressure features; "GPS" adds phone GPS speed.

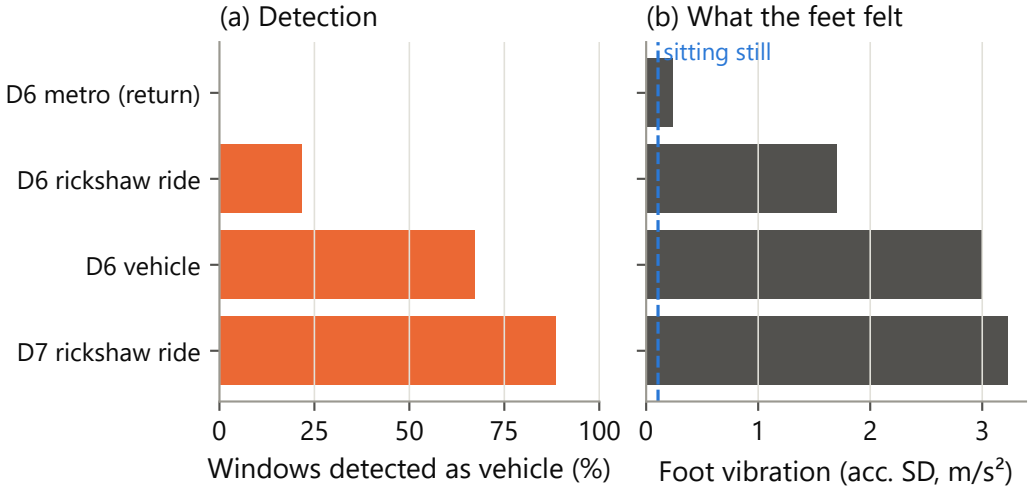



Figure 5: Vehicle detection by ride. (a) Share of each ride's windows recognised as vehicle (gradient boosting, smoothed, unseen day). (b) How much the feet vibrated (median acceleration SD); dashed: sitting still.

# 6 B. Place layer: where am I, from the air?

## Set-up

The 10-s environment data were averaged per minute (mean and spread of each variable) and given a 5-minute context (mean, spread and change), plus $PM_1/PM_{10}$ and $PM_{2.5}/PM_{10}$ ratios: 37 features. Labels came from the session contexts: 3,051 bedroom, 746 indoor, 377 outdoor and 49 vehicle minutes (vehicle only while riding; stations and markets unlabelled). Logistic regression, gradient boosting and a small CNN on the last 10 minutes of 10-s samples were compared, each with and without time of day. With time of day, "bedroom" becomes almost trivial at night; without it, the test is whether the air alone tells the place.

## Results

Table 5: Place recognition on unseen days (macro-F1, smoothed).

| Model | Air only | Air + time of day |
|---|---|---|
| Logistic regression | **0.67** | **0.68** |
| Gradient boosting | 0.60 | 0.63 |
| CNN (12k parameters) | 0.50 | 0.59 |

- The simplest model generalised best across days (Table 5). With about 4,200 minutes from six training days, flexible models fit day-specific details (the $CO_2$ sensor's low baseline on day 2, the hotter bedroom on some nights).
- Per class (logistic regression, air only): bedroom F1 0.96, indoor 0.77, outdoor 0.80, vehicle 0.15 (Fig. 6). Indoor minutes are sometimes taken for bedroom; most vehicle minutes for outdoor.
- *Time of day* adds little overall (0.67 → 0.68); it makes bedroom perfect but pushes indoor evening minutes towards outdoor. The air signature itself carries the information.
- *What the model uses* (Fig. 7b): sound level is by far the most informative, followed by humidity and $CO_2$. Particle levels hardly matter for place, because Dhaka's $PM_{2.5}$ is high everywhere.

- *Vehicles* (49 minutes, 44 of them on day 6) could not be learned: when day 6 is the test day, the training days contain one 5-minute rickshaw ride. The metro's strong signature ($CO_2$ to 1,500–2,500 ppm, a 6 °C temperature drop) is detected by the event layer (Section 8) but needs more rides to be learned as a place.

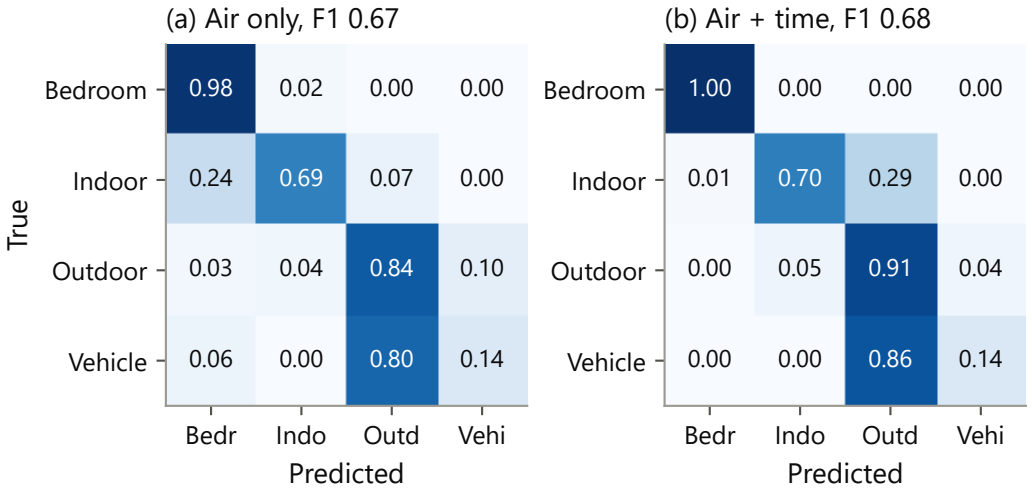



Figure 6: Place recognition with logistic regression on unseen days (smoothed), shown as the share of each true place (rows) assigned to each predicted place. (a) Environment node only; (b) with time of day added.

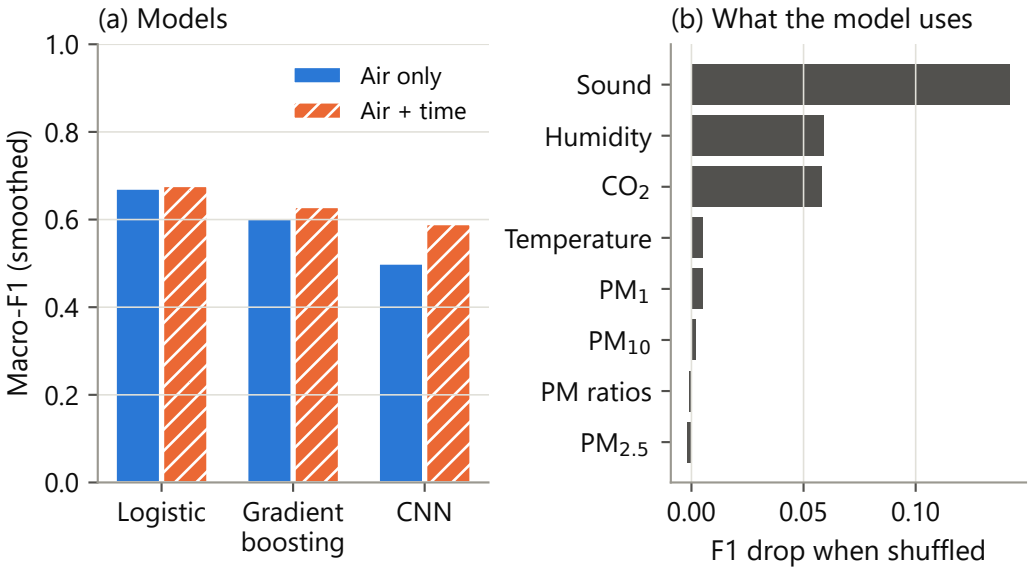



Figure 7: (a) Macro-F1 of the three place models with and without time of day (unseen days, smoothed). (b) Permutation importance (gradient boosting, air only): drop in macro-F1 when one variable's features are shuffled on the test day.

# 7 Fusion: the twin's context state

Place and activity were combined per minute into four twin states. The labels again came from sources independent of the inputs: bedroom (overnight session), indoor-still (indoor session and 0 band steps), outdoor-walking (walk session and $\geq$ 80 band steps) and vehicle (noted ride). Two fusion methods were compared: a *stacked* logistic regression on the place and activity probabilities (both produced by models that never saw the test day) plus GPS speed, and a *product of experts*, which treats air and feet as independent witnesses and multiplies the probability of each state's place and activity, without any training.

Fusion gave a modest overall gain (Table 6, Fig. 8). Its main benefit was for vehicles: recall rose from 0.14 (air) and 0.16 (feet) to 0.33 with the stacked model and GPS. Outdoor walking is recognised best by the feet and indoor sitting best by the air, as expected. Figure 9 shows the resulting twin for day 6, the most varied day: the night in the bedroom, the morning walk, rickshaw ride and indoor period, and the evening metro trip with its $CO_2$ peaks and the long period of high heart rate (the afternoon had no environment data).

Table 6: Twin-state recognition on unseen days (F1; 3,987 labelled minutes, 549 of them with insole data).

| Source | Bedroom | Indoor still | Outdoor walk | Vehicle | Macro-F1 |
|---|---|---|---|---|---|
| Place only (B) | 0.96 | 0.77 | 0.79 | 0.23 | 0.69 |
| Activity only (C) | 0.91 | 0.29 | 0.93 | 0.28 | 0.60 |
| Stacked B + C | 0.93 | 0.65 | 0.89 | 0.19 | 0.66 |
| Stacked B + C + GPS | 0.93 | 0.66 | 0.89 | **0.32** | **0.70** |
| Product of experts | 0.96 | 0.77 | 0.81 | 0.25 | **0.70** |

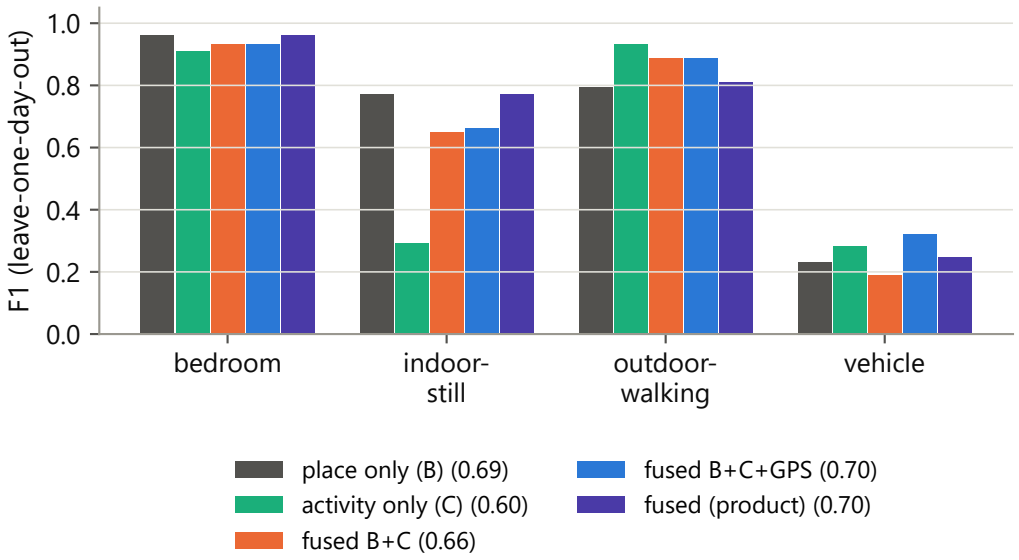



Figure 8: F1 of each twin state on unseen days for place only, activity only and three fusion variants (macro-F1 in brackets).

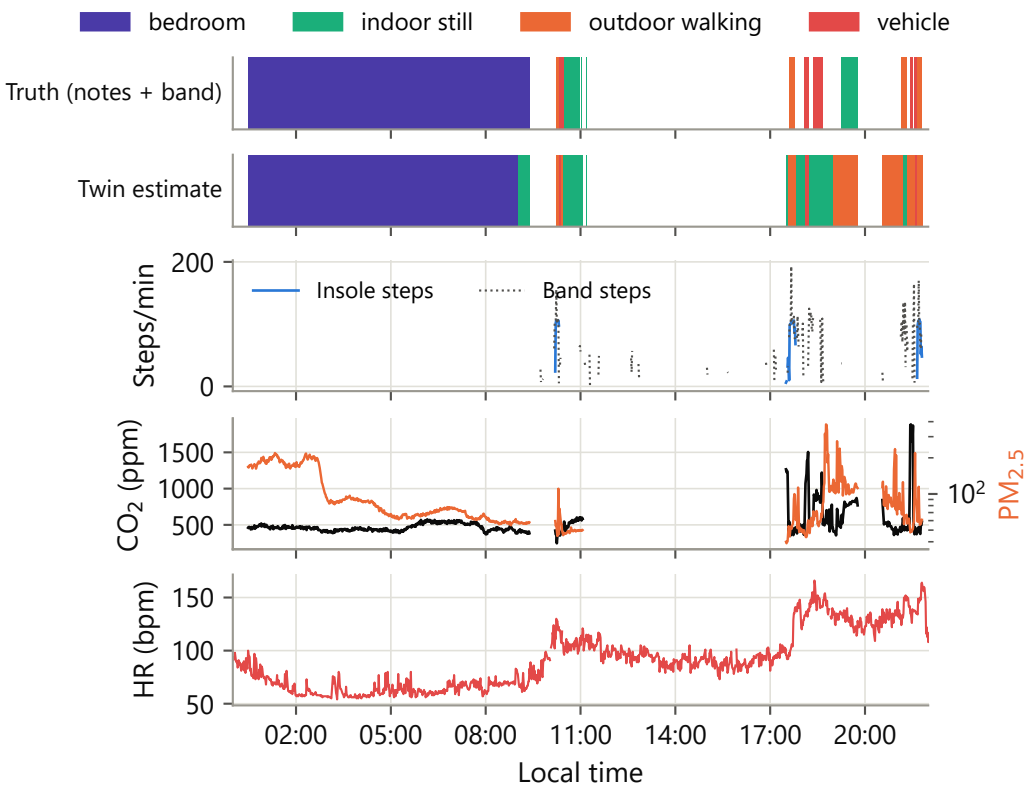



Figure 9: The digital twin of day 6 (13 October). From top: twin state from the participant's notes and the band (truth; blank = not labelled); the twin's estimate (stacked fusion with GPS, model not trained on day 6); step rate from the insoles and the band; $CO_2$ and $PM_{2.5}$ (log scale); wrist heart rate.

# 8 E. Event layer: slips and exposure spikes

## Slip detection

Every step in every walk (18,261 steps, 6.2 walking hours) was described by impact size, secondary peaks, maximum rotation speed, tilt range, duration and impact sharpness, each expressed relative to that walk's typical step (robust z-score, so that differences between walks and feet cancel). Two scores were compared.

- A *generic anomaly detector* (isolation forest on all features) ranked the slip 131st (top 0.7 %), which would mean 21 false alarms per walking hour at a threshold that catches it. Its most unusual steps were transitions: the first steps of walks, stops, stairs (day 4), boarding a rickshaw (day 6) and entering the restaurant (day 4). These are interesting gait moments, but not slips.
- A *physics-informed score*, the impact relative to the walk's typical step, ranked the slip **first** of 18,261 (6.1 robust standard deviations; Fig. 10). The margin is small, however: the next steps score 6.0 and 5.7 (hard heel strikes on other days), and 11 steps exceed 5.

With a single real slip no threshold can be validated. The result shows that the insole captures a slip as an extreme impact within an otherwise normal step, and that a slip detector should look for that specific signature rather than for any unusual step; it needs more (safely simulated) slips to be tuned.

## Exposure spikes

On the 10-s environment data a sample was flagged when $PM_{2.5}$ exceeded three times its trailing 10-minute median and that median plus 75 $\mu g/m^3$, or when $CO_2$ exceeded its trailing median by 400 ppm. Flagged samples were merged into episodes, each described by its peak, the fine-particle share at the peak ($PM_1/PM_{10}$), the $CO_2$ rise and the temperature change. Twenty episodes were found (fourteen overlapping the participant's notes); all seven noted exposure events were detected (Table 7).

The fingerprints separate the sources (Fig. 11): crowded, cooled vehicles raise $CO_2$ by 1,000–2,000 ppm with little change in particles and a 4–6 °C temperature drop; the soldering fume was entirely sub-micron; the smoke event combined coarse particles with a $CO_2$ jump, the signature of nearby

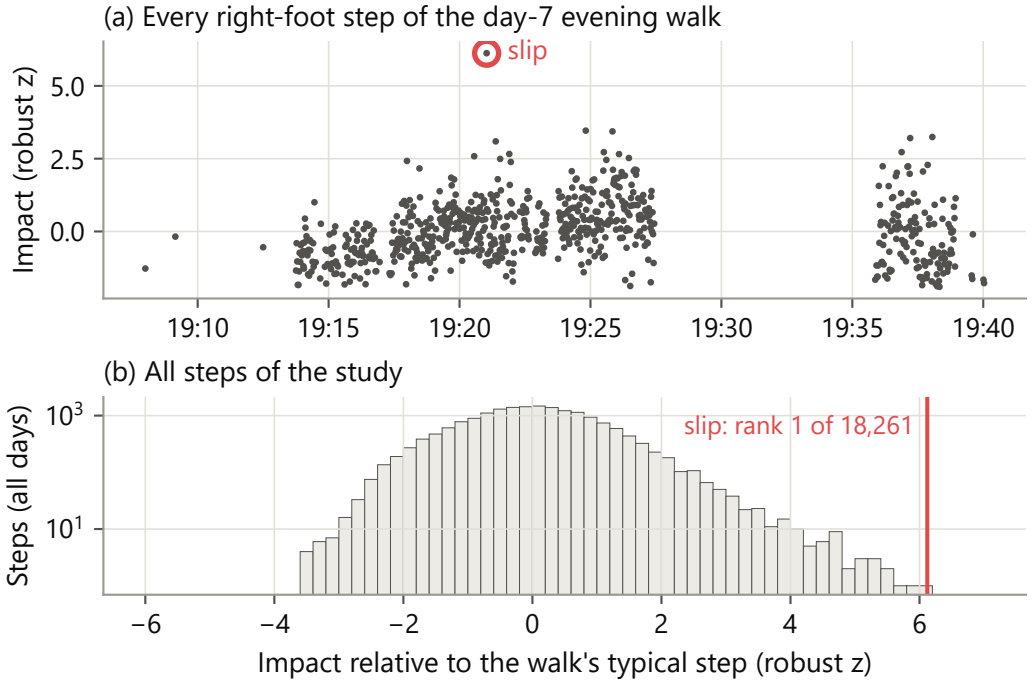



Figure 10: Slip detection without labels. (a) Impact of every right-foot step of the day-7 evening walk relative to that walk's typical step (robust z-score); circle: the participant's slip at 19:21:02. (b) The same score for all 18,261 steps of the study (log scale); the slip ranks first.

Table 7: Noted exposure events and the detected episodes.

| Day | Noted event | $PM_{2.5}$ peak ($\mu g/m^3$) | $CO_2$ rise (ppm) | Fingerprint |
|---|---|---|---|---|
| 4 | Dinner at a restaurant | 411 | 178 | mixed particles, some $CO_2$ |
| 6 | Metro train (outbound) | 63 | 1,064 | $CO_2$ only, temperature −5.7 °C |
| 6 | Metro station | 79 | 580 | $CO_2$ |
| 6 | Closed market | 514 | 429 | mixed particles |
| 6 | Smoke-filled space | 716 | 1,080 | coarse particles + $CO_2$ (combustion) |
| 6 | Metro train (return) | 51 | 1,998 | $CO_2$ only, temperature −4.4 °C |
| 7 | Soldering-iron fumes | 3,333* | 595 | 100 % sub-micron (flux fume) |

*sensor ceiling; the true peak was higher.

combustion. Of the six episodes without a note, four were short, coarse-particle peaks on outdoor walks ($PM_1$ share 0.18–0.32, typical of road dust; day 2 at 22:22 with 728 $\mu g/m^3$, and three peaks of 335–554 $\mu g/m^3$ on day 6 at 20:56–20:59), one a 20-second peak of 326 $\mu g/m^3$ on the day-4 morning walk, and one a $CO_2$ rise of 567 ppm on the same walk, probably breath close to the node. A phone twin could ask the wearer to label such episodes when they happen.

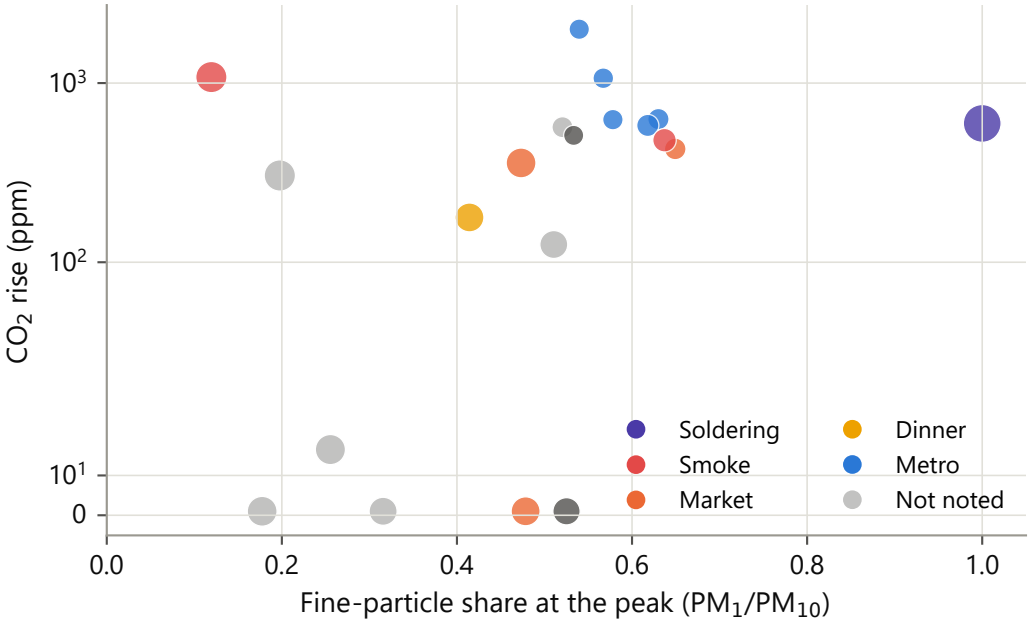



Figure 11: Fingerprints of the 20 detected exposure episodes: fine-particle share at the peak against $CO_2$ rise (symmetric log scale); marker size grows with peak $PM_{2.5}$; colour shows the participant's note (grey: no note).

# 9 Phone deployment

All models fit easily on a phone (Table 8). Processing one minute of both insoles costs about 30 ms of computation, well below 1 % of one core. The networks can be exported to TensorFlow Lite or ONNX and quantised to 8-bit; the gradient-boosting model can be shrunk (fewer trees) or compiled to code. The heaviest part on a phone will be the Bluetooth streaming and the orientation filter, not the models.

Table 8: Model size and speed. Time per call on one desktop CPU core with a batch of one, as a stand-in for a phone.

| Model | Input | Parameters | Size | ms/call |
|---|---|---|---|---|
| Heel-strike TCN | $4 \times 1000$ (10 s) | 32,385 | 32 kB | 2.0 |
| Toe-off CNN | $4 \times 100$ (stride) | 34,065 | 33 kB | 0.2 |
| Activity CNN | $10 \times 500$ (10 s) | 50,291 | 49 kB | 0.5 |
| Activity gradient boosting | 37 features | – | 3.2 MB† | 8.0 |
| Place logistic regression | 37 features | – | 4 kB† | 0.9 |

Network sizes as 8-bit weights; †serialised Python object.

# 10 Discussion

## What the experiments teach

1. **Labels limit accuracy more than models do.** The toe-off models hit the pressure reference's own noise floor; correcting the vehicle labels did more for activity recognition (+0.08 macro-F1) than any model change; missed heel strikes were mostly steps without a clear impact.

2. **Independent labels prevent circular results.** Using the wrist band for walking labels, the participant's notes for rides and places, and the IMU only as input, ensures that the models learn the world rather than the labelling rule.

3. **Simple models generalise better with little data.** Logistic regression won for place, gradient boosting on engineered features for activity; the neural networks only won where examples were plentiful (heel strikes, about 19,000 impacts).

4. **Physics beats complexity where it applies.** The two-foot alternation check halved false steps in vehicles; impact-relative-to-own-steps found the slip where a generic anomaly detector did not; particle size and $CO_2$ separated fume, smoke and crowding.

5. **Layers cover each other's blind spots.** The feet cannot feel a smooth metro ride but the air shows it; the air cannot tell sitting from walking outdoors but the feet can. Fusion improved vehicle recall most.

6. **Sensor robustness is part of the twin.** Pressure sensors failed or were replaced three times; an IMU-based design and one-insole operation (macro-F1 0.74–0.77) keep the twin working.

### Limitations

- One participant and seven days: all results describe this person and this city. Days are not fully independent (same home, same insoles).
- Vehicles: 23 minutes of insole data and 49 environment minutes, on two days, are too few to learn rides reliably.
- One slip: the slip result is a case study, not a validated detector.
- The labels are imperfect: the pressure toe-off shifts between sessions, the band undercounts steps, and session contexts are coarse (walks begin indoors).
- The neural networks were trained once (single random seed); their scores vary by a few hundredths between runs.
- Models were run on a desktop CPU; on-phone latency and battery use remain to be measured.

### Next: the second participant

The second participant allows the step from "a twin of one person" to "a twin that transfers":

- *Leave-one-person-out tests* of every model trained on this participant, which is the real test of the mounting- and body-independent design.
- *Personalisation*: how many of her own labelled minutes are needed to bring each model to its within-person accuracy.
- *Protocol*: the same sessions; notes of every ride, stairs and indoor/outdoor change (her labels); a known test pour on the bottle; ideally a few rides by rickshaw and metro and a short stair bout, which this dataset lacks most.

## 11 Conclusion

A phone-sized machine-learning layer was built on top of the self-monitoring data and tested strictly on unseen days. The gait layer finds heel strikes from the foot IMU alone (F1 0.92, timing within about 9 ms) and shows that the IMU toe-off rule is as precise as its pressure reference allows. The

activity layer recognises walking and sitting from either insole (F1 0.91), the place layer recognises bedroom, indoor and outdoor from the air alone (F1 0.77–0.96), and their fusion forms the twin's context state. The event layer catches every noted exposure episode with an interpretable fingerprint and isolates the participant's slip by its impact. Vehicles and slips remain limited by the number of examples, which the second participant and targeted recordings can address.

# 1 Reproducing the results

Run from `pipeline/` after the daily analyses, `compare_days.py`, `relationships.py` and `walk_hydration.py`:

1. `python ml_gait.py .. -rebuild` (toe-off dataset and models; `ml/A_gait/`)
2. `python ml_heelstrike.py .. -rebuild` (heel-strike TCN)
3. `python ml_activity.py .. -rebuild` (activity; `ml/C_activity/`)
4. `python ml_place.py .. -rebuild` (place; `ml/B_place/`)
5. `python ml_fusion.py ..` (twin states; `ml/fusion/`)
6. `python ml_events.py ..` (slips and exposure episodes; `ml/E_events/`)
7. `python ml_figures_a.py ..` (gait figures and `ml/deploy.json`)

All figures are written to `ml/figures/` as PDF and PNG; numerical results are in the `results.json` file of each task folder.